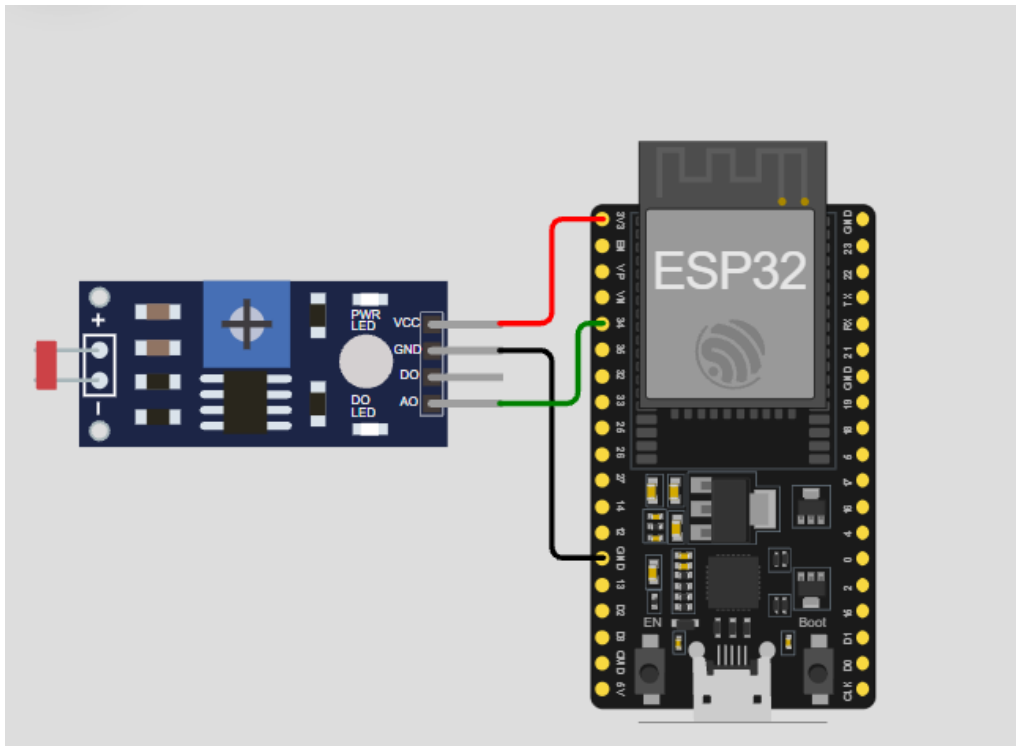


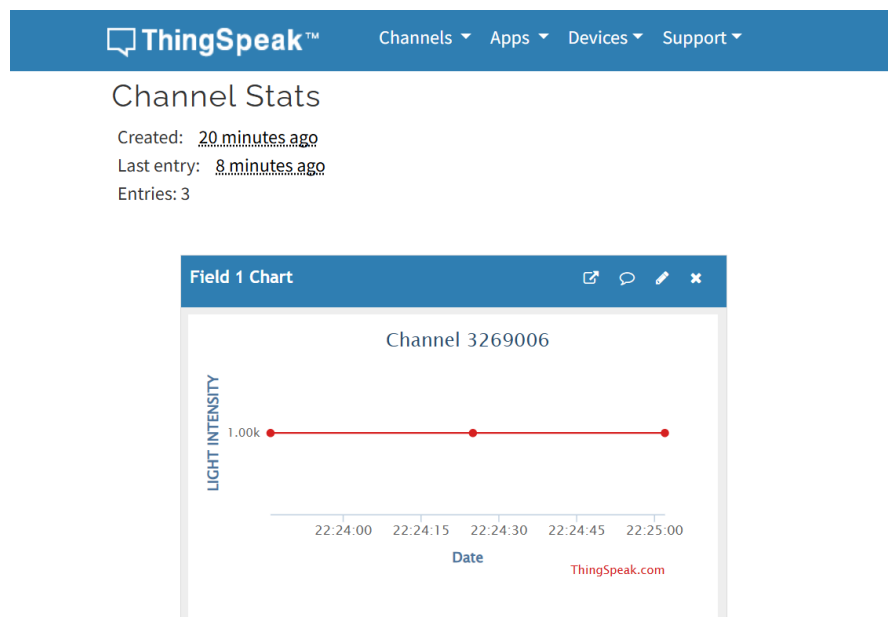
Embedded System and Internet of Things

Task:-04

1. Circuit Diagram:-



6



2. Source Code:-

```
#include <WiFi.h>

#include "ThingSpeak.h"

const char* ssid = "Wokwi-GUEST";

const char* password = "";

WiFiClient client;

unsigned long myChannelNumber = 3269006;

const char * myWriteAPIKey = "2EL9OEBOM9I12X0R";

int ldrPin = 34;

void setup() {

  Serial.begin(115200);

  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

  }
```

```
Serial.println("WiFi Connected");

ThingSpeak.begin(client);

}

void loop() {

  int ldrValue = analogRead(ldrPin);

  Serial.print("Light Value: ");
  Serial.println(ldrValue);

  ThingSpeak.setField(1, ldrValue);

  int x = ThingSpeak.writeFields(myChannelNumber,
myWriteAPIKey);

  if(x == 200){

    Serial.println("Data sent to ThingSpeak.");

  } else{

    Serial.println("Error sending data.");

  }

  delay(15000); // ThingSpeak limit (15 sec)
```

}

3. Explanation:-

A) How Device Sends Data to Cloud:-

Step 1 – Sensor Reading

The ESP32 reads light intensity using LDR connected to analog pin GPIO 34.

Step 2 – WiFi Connection

ESP32 connects to internet using built-in WiFi.

Step 3 – Cloud Communication

The sensor value is sent to ThingSpeak using:

- Channel ID
- Write API Key
- HTTP API request

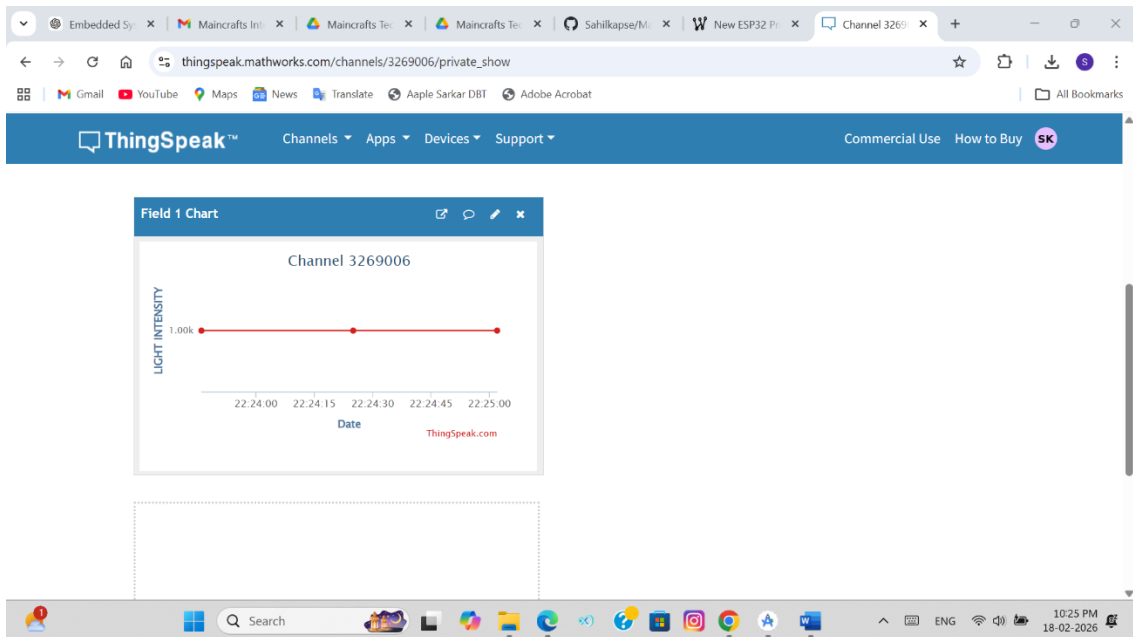
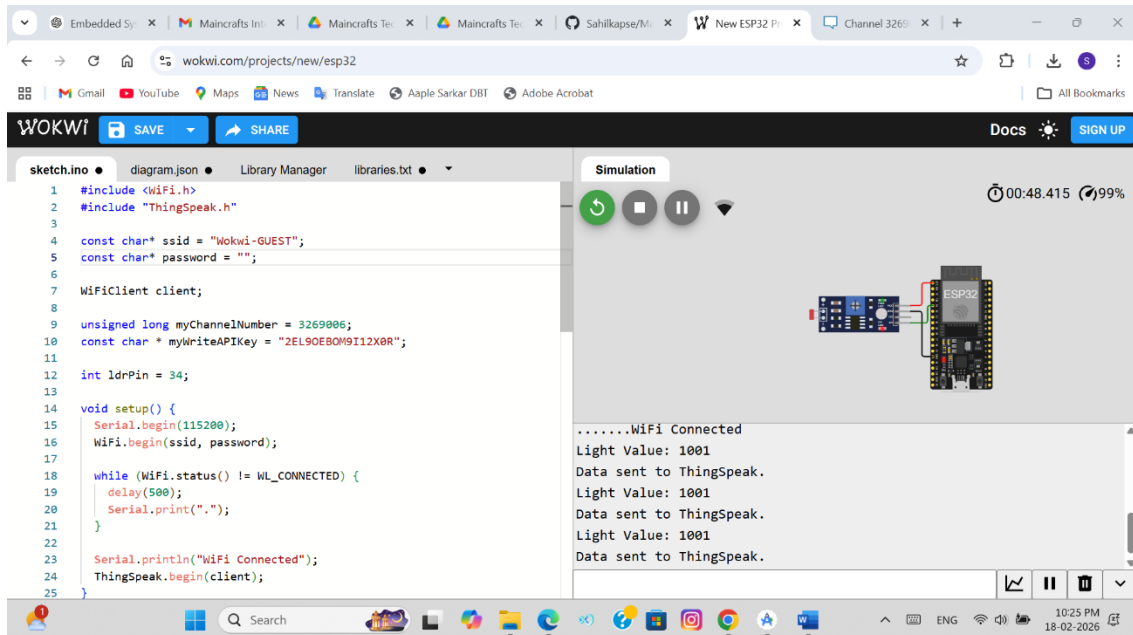
Step 4 – Dashboard Display

ThingSpeak displays:

- Live light reading
- Graph chart
- Historical data

This demonstrates **real-time IoT device-to-cloud communication**

4. Output:-



Cloud Dashboard Link:-

<https://thingspeak.mathworks.com/channels/3269006/edit>